Look Both Ways: Reconstructing Two Hidden Months of Post-Earthquake Mobility

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Abstract

The HuMob Challenge 2026 withholds the daily origin-destination flows of February and March 2024, the second and third months after the Noto Peninsula earthquake, and releases the months on both sides. We contribute a model that reconstructs this region-wide blackout from the data on both sides, and a way to validate it. The model averages three estimates: a two-anchor interpolation whose within-cell path follows published post-disaster recovery curves, a low-rank state-space smoother fitted across both observed periods, and TimesFM-3 forecasts of within-cell flows run forward and backward. We chose the components on validation windows cut from the observed months to reproduce the gap's length and anchors. On May to June the model's error is 8.4% below that of the following month's mean, and on nine windows shifted a week from our rolling validation windows, which no choice used, it is 7.5% below. In an ablation on these windows each estimate lowers the error, and on the rolling windows two established completion methods, Bayesian temporal matrix factorization and per-pair linear interpolation, do worse than the model. On windows that contain the earthquake or start just after it, three design choices, among them the smoother, raise the error instead, so such a model has to be checked on both kinds of window. From late January to April, the observed data show within-cell flows rising by 26% in Wajima and 18% in Anamizu as a share of all flows.

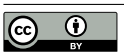
CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → *Machine learning*.

Keywords

human mobility, origin-destination flows, disaster recovery, missing data, time-series foundation models, model validation

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1 Introduction

The HuMob Challenge 2026 asks how far human mobility recovered after the 2024 Noto Peninsula earthquake, using data that stop four weeks after the earthquake and resume two months later [27, 28]. Daily origin-destination (OD) flows between grid cells are released for the months on both sides, and those of February and March 2024 must be reconstructed. These were the second and third months after the M7.6 earthquake of January 1, when displacement, repair and return were under way. Reconstructing them estimates the course of the recovery. Mobility studies of this earthquake mostly analyze evacuation in its first hours and days [9, 16], and one tracks monthly visitors to a single city [22].

The task is an interpolation across a region-wide data blackout, and none of the three nearest method families addresses it directly. Methods for mobility after disasters estimate displacement from observations made during and after the event, often online [2, 15, 25], whereas here nothing arrives for two months. Spatiotemporal imputation fills gaps from locations that remain observed [6, 14, 17]. On traffic blackouts it has been evaluated on gaps of one hour with every sensor dark [4] and of up to two hours at single sensors [21], against 58 scored days here. Pretrained time-series models extend a series from one side of a gap, and those built for imputation have been evaluated on interior gaps of up to four days [13]. In past HuMob challenges, individual trajectories were predicted [8, 23].

We reconstruct the test period from both sides by averaging three estimates: a two-anchor interpolation whose within-cell path follows published recovery curves, a low-rank state-space smoother fitted across both observed periods, and TimesFM-3 forecasts of within-cell flows run forward and backward. We chose the components on validation windows cut from the observed months to reproduce the gap's length and anchors. On May to June the model's error is 8.4% below that of the following month's mean, and on nine windows shifted a week from our rolling validation windows, which no choice used, it is 7.5% below. On the rolling windows, two established completion methods, Bayesian temporal matrix factorization and per-pair linear interpolation, do worse. Three design choices, the smoother, fitting on the evaluation region and joint forecasting, help on windows shaped like the test period and hurt on windows that contain the earthquake or start just after it. Between late January and April, within-cell flows rose by 26% in Wajima as a share of all flows.

2 Task and Data

The challenge data are daily OD flows between cells of about 2 km on a 100 by 70 grid over the Noto region of Ishikawa Prefecture [27, 28]. The dataset of an earlier HuMob challenge is described by Yabe et al. [29]. A between-cell (off-diagonal) flow counts people

moving from one cell to another on a day, and a within-cell (diagonal) flow counts people who remain in the cell. The counts are anonymized and scaled by a constant [27], and each day's flows, including a bucket for people outside the grid, sum to the same total. The flows are therefore shares of a fixed total. Flows are observed from November 1, 2023 to January 31, 2024, which includes the earthquake, and from April 1 to October 31, 2024, with 14 of the 306 days in the file missing. The task is to predict every OD pair on each day from February 1 to March 31, 2024, of which 58 days are scored.

Scoring is restricted to an evaluation region of 1,476 cells, in which only 1,452 pairs (922 cells and 530 between-cell pairs) carry flow on any observed day. For each day the score takes the root-mean-square error over the 1,476 within-cell flows and over the 2,177,100 between-cell pairs, with missing pairs counted as zero. It divides each by a fixed constant (26.57 for the diagonal, 0.0176 for the off-diagonal) and averages the two normalized errors (NRMSE). The combined score averages this over days. Two baselines are reported for the test period: the April 2024 mean of each pair (0.2261) and its November to December 2023 mean (0.2800) [27].

3 Method

We predict every OD pair of the evaluation region on every scored day of the test period from three estimates, each of which uses the observations on both sides of the gap (Figure 1b).

3.1 Two-Anchor Interpolation

We give each OD pair a level from each of two anchor windows, the before-anchor (the ten days before the gap) and the after-anchor (the month after it). For the test period these are January 22 to 31, eight of them observed, and April 1 to 30. Each anchor day is first divided by the pair's day-of-week factor, so that a level does not depend on the weekdays its window holds. The factors are estimated per pair on the observed days after the gap (in a validation window, on those from April onward outside its target) and shrunk toward the flow-weighted weekly shape of the evaluation region. On the diagonal, empirical Bayes (EB) sets the shrinkage weight of each cell and weekday, $\lambda = s^2/(s^2 + \sigma_b^2)$, where s^2 is the sampling variance of the cell's factor and σ_b^2 the flow-weighted variance of the factors across pairs, net of sampling variance. Between-cell pairs rest on few non-zero days, and for them the weight is fixed at 0.35. Weekday holidays outside Golden Week and Obon, the New Year days included, share one pooled factor, the ratio of their total flow to that of the same weekday. Three of them fall in the test period.

On gap day t, with $t = 0$ on February 1, we predict pair c as

$$\hat{x}_{c,t} = \left[w(t)\, L_c^{\mathrm{b}} + \big(1 - w(t)\big)\, L_c^{\mathrm{a}}\right] f_{c,k(t)}\, h_t, \tag{1}$$

where L_c^{b} and L_c^{a} are the anchor levels, $f_{c,k}$ is the factor of day type k, h_t is the holiday factor, and the weight $w(t)$ encodes the recovery path over the T days from the start of the gap to the first after-anchor day. Within-cell flows count people who stay in a cell. They fall as residents are displaced and recover as they return, and the time course of that return after disasters has been measured directly. For these flows w is the normalized exponential curve $w_\tau(t) = (e^{-t/\tau} - e^{-T/\tau})/(1 - e^{-T/\tau})$, averaged over a flat prior on τ spanning the time constants Yabe et al. [26] fitted to displacement after the 2016 Kumamoto earthquake (7.6 days) and after Hurricane Maria (26.9 days), where restoring power took over 200 days. Between cells, $w = 1 - \sqrt{t/(T-1)}$ moves most early on.

3.2 Low-Rank State-Space Smoother

In the smoother, the path is estimated from both observed periods instead of being fixed. Every observed day is divided by day-of-week factors, shrunk by the fixed weight and estimated on all observed days but December 29 to January 5. Each pair's mean is then subtracted, and the pairs-by-days matrix is projected onto its 12 leading principal components. The coordinates of each component follow a local linear trend whose level, slope and observation noise variances, as multiples of the coordinate's observed variance, are chosen by maximum likelihood from $\{10^{-4}, \ldots, 10^{-1}\}$, $\{10^{-6}, \ldots, 10^{-3}\}$ and $\{0.1, 0.5, 1\}$. A Kalman filter is run forward over the calendar, skipping the update on unobserved days, and a Rauch–Tung–Striebel smoother [20] is run backward, which carries information from after the gap into it. The second estimate is obtained by mapping the smoothed coordinates back, restoring the day-of-week factors and clipping to between zero and three times each pair's largest observed value. The dynamics matter: a low-rank model without them cannot recover a slice with no observations [19].

3.3 Pretrained Forecasts for Within-Cell Flows

For each cell with non-zero flow in either anchor window, we forecast the within-cell series with TimesFM-3 [7, 12], forward from the whole observed block before the gap (92 days from November 1, missing days interpolated linearly) and backward from the time-reversed block after it (214 days from April 1). On gap day t we average the forward and backward forecasts $\hat{y}_t^{\mathrm{f}}$ and $\hat{y}_t^{\mathrm{b}}$ as in forward and backward forecasting ensembles [18], but weighted toward the nearer edge: $(b_t \hat{y}_t^{\mathrm{f}} + f_t \hat{y}_t^{\mathrm{b}})/(f_t + b_t)$. Here f_t and b_t count the days since the last observed day before the gap and until the first after it. We use the model zero-shot with its default settings, one series at a time, in its MLX build. We do not forecast between-cell pairs. They are sparse count series, and on the intermittent series of a recent benchmark the best methods were classical [11], as in Section 5.2.

3.4 Combination and Fitting Region

We average the interpolation and the smoother with equal weights for every OD pair, and on the diagonal we average the result with the forecast, again with equal weights. A cell with no flow in either anchor window has no forecast, which counts as zero, and receives half of the first average. We fit all pooled quantities (the weekly shape toward which factors shrink, the holiday factor, and the smoother's components and day-of-week factors) on the pairs of the evaluation region alone, which we call the region fit. Fitted on the whole grid, they would also describe the 5,524 cells outside the region, and Section 5.3 measures the difference. Pairs with no flow on any observed day are predicted as zero.

4 Validation

Because the test period is never observed, we chose the components on validation windows cut from the observed periods. Each window hides a target span and keeps anchor windows that play the roles of

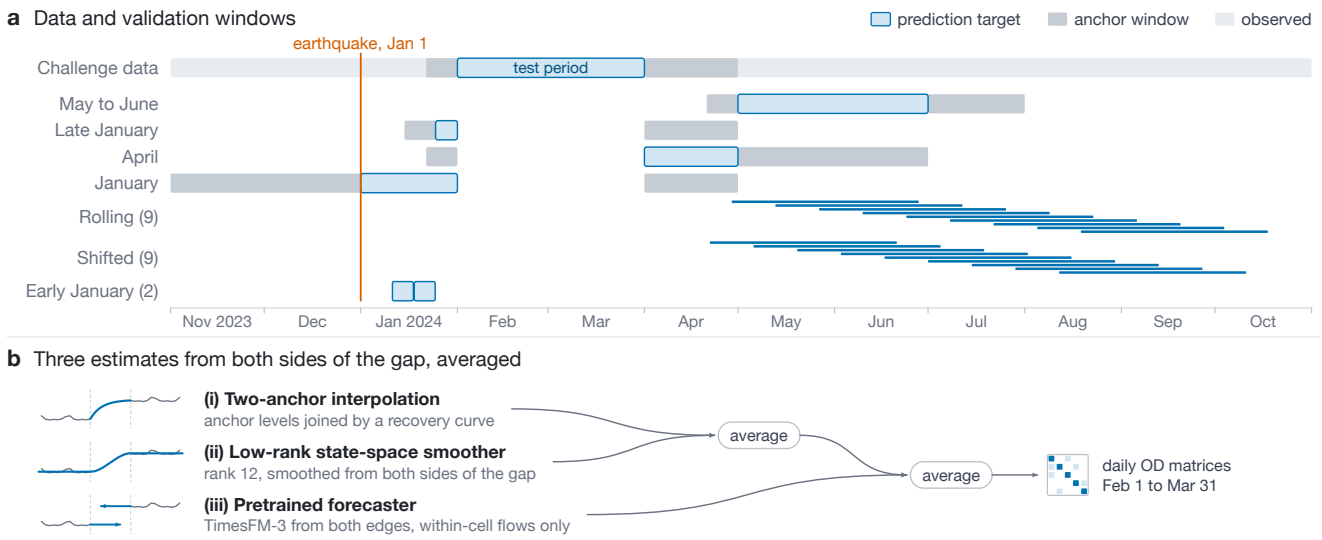


Figure 1: (a) Data and validation windows: each target (blue) is predicted from anchor windows (darker grey; named windows only); no choice used the shifted windows. (b) The model averages (i) and (ii) for every pair, then (iii) on the diagonal.

Table 1: Ablation: combined score (lower is better; bold: best) as components are added. Last row: smoother row fitted on the whole grid. Rolling, Shifted: nine-window means.

Model	May–Jun	Late Jan	April	January	Rolling	Shifted
After-anchor mean	0.2214	0.2466	0.2161	0.3177	0.2238	0.2237
Interpolation	0.2062	0.2229	0.2090	**0.3094**	0.2095	0.2094
+ smoother	0.2037	0.2095	0.2049	0.3350	0.2075	0.2078
+ TimesFM-3	0.2030	**0.2054**	0.2032	0.3365	0.2068	0.2070
+ EB factors	**0.2029**	0.2055	**0.2030**	0.3365	**0.2067**	**0.2069**
Whole-grid fit	0.2059	0.2158	0.2071	0.3243	0.2082	0.2085

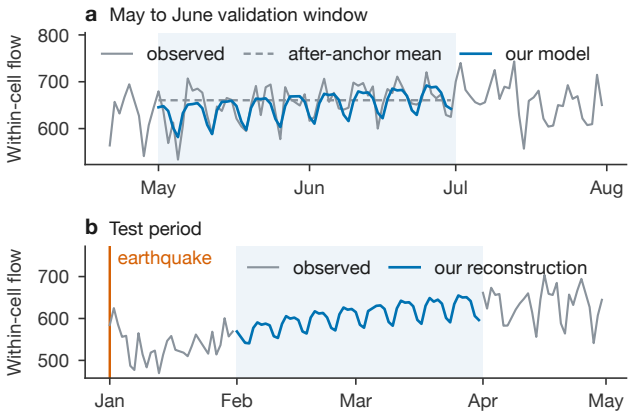


Figure 2: Within-cell flow of the cell that contains central Nanao. (a) Validation window from May to June. (b) Reconstruction of the test period.

late January and April (Figure 1a). Work on spatial prediction holds that validation should match the conditions of deployment [3], and we apply this to time by reproducing the gap's length and anchors.

- *May to June* has about the length of the test period and anchors in late April and July.
- The *edge windows*, late January (January 25 to 31) and April, each share one of the test period's anchors.
- *January* contains the earthquake, a month before the test period begins, and served only as a secondary check.
- *Nine rolling windows* of about 60 days, starting every two weeks from April 29, take the ten days before and up to 30 days after each target as anchors. *Nine shifted windows*, built the same way a week earlier, entered no choice.
- *Two early-January windows* (January 11 to 17 and 18 to 24, the second with four observed days) start forward forecasts 10 and 17 days after the earthquake, against 31 days for the test period.

The forecasts use the observed block on each side of a window's target. Their contexts therefore differ from the test period's 92 and 214 days: 30 and 123 days on May to June, and 14 to 140 days on the rolling windows. The curve runs over each window's target days, as in the gap wherever both anchors touch the target. Sections 5.1 and 5.3 also place it where the edge windows and January would sit in the gap. We compare two models by their mean per-day difference in combined score, with 95% intervals from a paired bootstrap (4,000 resamples) over days, or dates for pooled windows. For later decisions, such as the forecaster (Section 5.2) and the public-records test (Section 5.4), we fixed the rule in advance.

5 Results

5.1 Ablation Study

In Table 1 we add the components one at a time. The interpolation lowers the error of the after-anchor mean, the validation counterpart of the April-mean baseline, by 6.9% on May to June and 6.4% on

the rolling windows. Its day-of-week and holiday factors alone give 5.1% and 4.6%, and its before-anchor and recovery curve a further 1.9% on both (95% intervals 1.2% to 2.5% and 1.6% to 2.1%). With the smoother, the error is lowered by a further 1.2% and 0.9%. The full model scores 0.2029 on May to June and 0.2067 on the rolling windows, 8.4% and 7.6% below the after-anchor mean. On May to June it lowers the within-cell NRMSE from 0.104 to 0.087 and the between-cell NRMSE from 0.339 to 0.319. On the nine shifted windows, which no choice used, it is 7.5% below the after-anchor mean (95% interval 6.5% to 8.5%) and lower on all nine, and every component keeps its sign with an interval that excludes zero.

Established completion methods, which we ran on the same windows, do worse. On the rolling windows, per-pair linear interpolation between the two anchor means scores 0.2215, below the after-anchor mean (0.2238), and Bayesian temporal matrix factorization (BTMF) [5], with its rank chosen on May to June, scores 0.2263, above it. Our model is 6.7% and 8.6% below them (95% intervals 5.8% to 7.5% and 7.3% to 10.1%). The model's constants matter little: smoother ranks from 4 to 24, a smoother weight of 0.7 instead of 0.5 and a single time constant at either end of the prior change the rolling-window error by at most 0.06% either way. A smoother weight of 0.3 and forecast weights of 0.25 and 1.0 on the diagonal raise it by 0.20%, 0.06% and 0.24%.

On the edge windows the curve, run over the target alone, falls in one week on late January instead of over the first week of a two-month path. Placed where these windows would sit in the gap, the before-anchor and curve lower the error by 11.3% on late January and 0.7% on April (95% intervals 9.0% to 13.7% and 0.5% to 1.0%). Every step in Table 1 then keeps its sign, and the smoother's and the forecaster's gains shrink (late January: from 6.0% to 2.5% and from 1.9% to 0.4%; April: from 2.0% to 0.7% and from 0.8% to 0.3%). However, on January, which contains the earthquake, the interpolation alone scores best, and the full model trails the after-anchor mean by 5.9% (95% interval 1.9% to 9.6%). In Figure 2 the model follows the weekly cycle of one large cell on May to June and, for the test period, rises from its January toward its April level.

5.2 The Pretrained Forecaster

TimesFM-3 helps on the within-cell flows only. Used alone for every pair with flow in either anchor window, it scores 0.2168 on May to June against 0.2037 for the interpolation and smoother combined. Nearly all of the difference is between cells (off-diagonal NRMSE 0.3445 against 0.3188, diagonal 0.0891 against 0.0886). On the diagonal alone, averaged with that combination, it lowers the error by 0.33% on May to June and 0.31% on the rolling windows (95% intervals 0.17% to 0.49% and 0.23% to 0.40%), on seven of the nine rolling windows. Simpler within-cell estimates in its place (the after-anchor mean, linear or square-root interpolation) raise the error on May to June by 0.10% to 1.34%. Shorter contexts only raise its error. Cutting the forward context to its last 14 or 28 days costs 0.47% and 0.16% on the rolling windows (95% intervals 0.37% to 0.57% and 0.11% to 0.21%) and cutting the backward context to 30 days costs 0.10%. Cuts to 56 days forward or 60 and 120 days backward change it by less than 0.03%. However, its own uncertainty does not indicate when it helps. On rolling-window cells with a mean anchor level of at least 20, the relative width of its 80% interval separates the cell-days it wins from the rest no better than chance. The area under the ROC curve is 0.495 (95% interval over dates 0.488 to 0.503).

We tested other forecasters in its place against a rule fixed in advance. It required a pooled rolling gain of at least 0.05% with a 99% interval (Bonferroni over five variants) excluding zero, a gain on May to June, and no loss at the 95% level on May to June, late January or April. Chronos-2 [1] gained on May to June and the edge windows, but its pooled rolling gain (0.06%) has an interval containing zero. The mean of TimesFM-3, Chronos-2 and PatchTST-FM-r2 [24] gained 0.04%, below the floor, and PatchTST-FM-r2 lost 0.06%. Run as a masked reconstruction that fills the gap from both edges, PatchTST-FM-r2 lost 4.53% on late January, where the levels on either side of the gap are far apart. Only joint forecasting of the 32 largest cells [12] passed, with a pooled rolling gain of 0.10%. A second rule fixed in advance, no loss on the two early-January windows, rejected it: over their 11 scored days it raised the error by 0.69% (95% interval excluding zero), mostly through its forward forecasts. Because the test period's forward forecasts start from a January context with the earthquake, we forecast one cell at a time.

5.3 Dependence on the Validation Window

Three design choices that help on the windows shaped like the test period hurt on windows that contain the earthquake or forecast from just after it, each time with 95% intervals that exclude zero on both sides. The region fit lowers the error by 1.07% on May to June and 2.94% on late January but raises it by 3.29% on January, and 88% of its May to June gain comes from fitting the smoother on the region. The smoother lowers the error by 1.2% on May to June and raises it by 8.3% on January, and joint forecasting lowers it on the rolling windows and raises it on the early-January windows. Both January reversals hold with the curve placed as in the gap (the region fit and the smoother raise the error by 3.6% and 7.9%). The interpolation's and the forecaster's effects on January change sign with that placement, from a 2.6% gain to a 0.5% loss and from a 0.47% loss to a 0.41% gain. We resolved each reversal by the windows that match the component's position in the gap. Because the test period begins a month after the earthquake, the smoother and region fit stay, and because its forward forecasts start from a January context with the earthquake, joint forecasting goes.

5.4 Recovery by Municipality

In the prefecture's disaster bulletins [10], the people in secondary evacuation at inns and hotels are counted: 4,792 on January 31, 5,275 on February 16 and 3,312 on April 2. The interpolation, monotone between its anchors, cannot follow such a peak. The bulletins also give each municipality's shelter evacuees, which fall from late January to April. Wherever they cover both anchors, we tested taking each cell's interpolation path from its municipality's series, scaled between the anchors. This raises the error by 0.06% on May to June (95% interval 0.02% to 0.09%) and by 0.01% on the rolling windows (0.004% to 0.02%; the records reach four of them). Averaging that path with the recovery curve roughly halves both losses, and the gains on the edge windows have intervals that include zero.

Because both anchors are observed, we can measure how much each municipality changed without the model. Each cell is assigned

to the nearest municipal centre. From the before-anchor (January 22 to 31) to April, the within-cell flows of a municipality's cells, as a share of all flows that day, rose by 26% in Wajima (95% interval 20% to 32%) and 18% in Anamizu (14% to 23%). They rose by 4% to 9% in Hakui, Nanao, Shika and the town of Noto, close to the 7% rise of the flows outside the grid, and fell by 3% in Nakanoto. The interval for Suzu includes no change. In the six municipalities whose share clearly rose, the reconstruction's weekly mean reaches halfway between February 19 and March 12, one to four weeks after the recovery curve alone (February 12). The data on both sides fix how far each share moved, and the model sets the timing inside the gap.

6 Conclusion

Averaging three estimates, a two-anchor interpolation, a low-rank state-space smoother and bidirectional TimesFM-3 forecasts, lowers the validation error of the after-anchor mean by 8.4% on May to June and 7.5% on nine windows shifted a week from the rolling ones. Results can reverse between windows near the earthquake and away from it, and we resolved each by the windows that match a component's position in the gap. A reconstruction across a gap has to look both ways, at the data on both sides and at both kinds of window. The two sides measure how much each municipality changed, with Wajima's within-cell share up 26%. The timing inside the gap comes from the model. Code for the model and the experiments: https://github.com/mustafasameen/humob2026-noto.

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